

Project Title & Team Details :-

MetroMitra

An AI-Powered Smart Metro Assistant: From Complaint

Chaos to Governance Clarity

Track no:-05 (AI For Society)

Team Name & Members:-



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National Innovation Hackathon 2026



Problem Identification & Relevance:-

The Problem



- Manual grievance handling is slow and inefficient.
- Commuters face difficulty reporting issues.
- Lack of real-time updates and complaint tracking.
- Metro authorities receive scattered feedback from multiple channels.

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Alignment (AI for Society)

- Improves public service delivery using AI.
- Enables transparent and accountable governance.
- Enhances commuter experience through digital assistance.



Relevance

- Designed for Patna Metro and adaptable to other metro systems.
- Reduces complaint resolution time.
- Supports smarter, citizen-centric urban transportation.

Innovation & Originality :-

What current grievance handling misses, and what MetroMitra adds



Scattered Complaints

Complaints arrive from multiple channels are hard to unify.



Language Diversity

Users report in Hindi English, and mixed local language.



Slow Manual Review

Human storing delays action on urgent complaints.



Duplicate Issues

Repeated complaints are hndled separately instead of as one pattern.

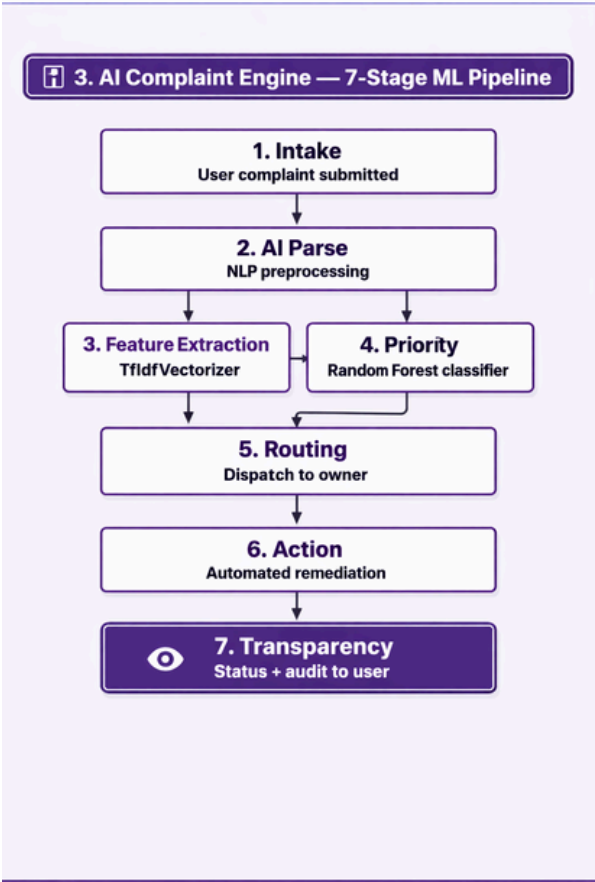
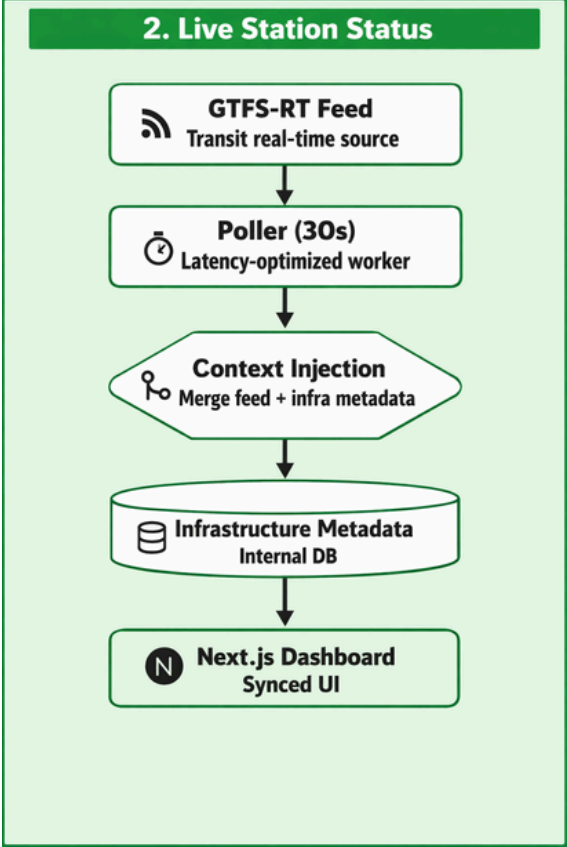
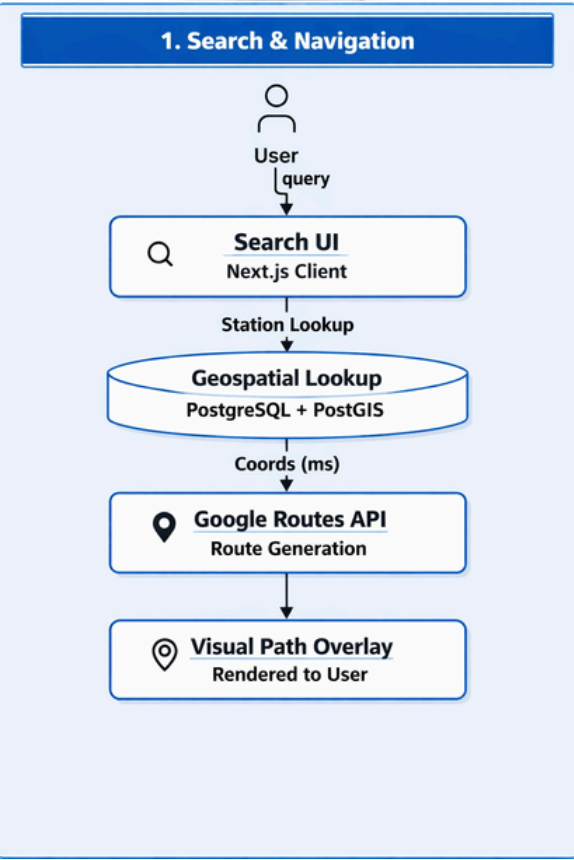


No Priority Intelligence

Authorities cannot instantly see which issue needs action first.

From complaint noise to actionable governance intelligence.

Solution Design & Technical Architecture :-



Frontend

Backend

Database

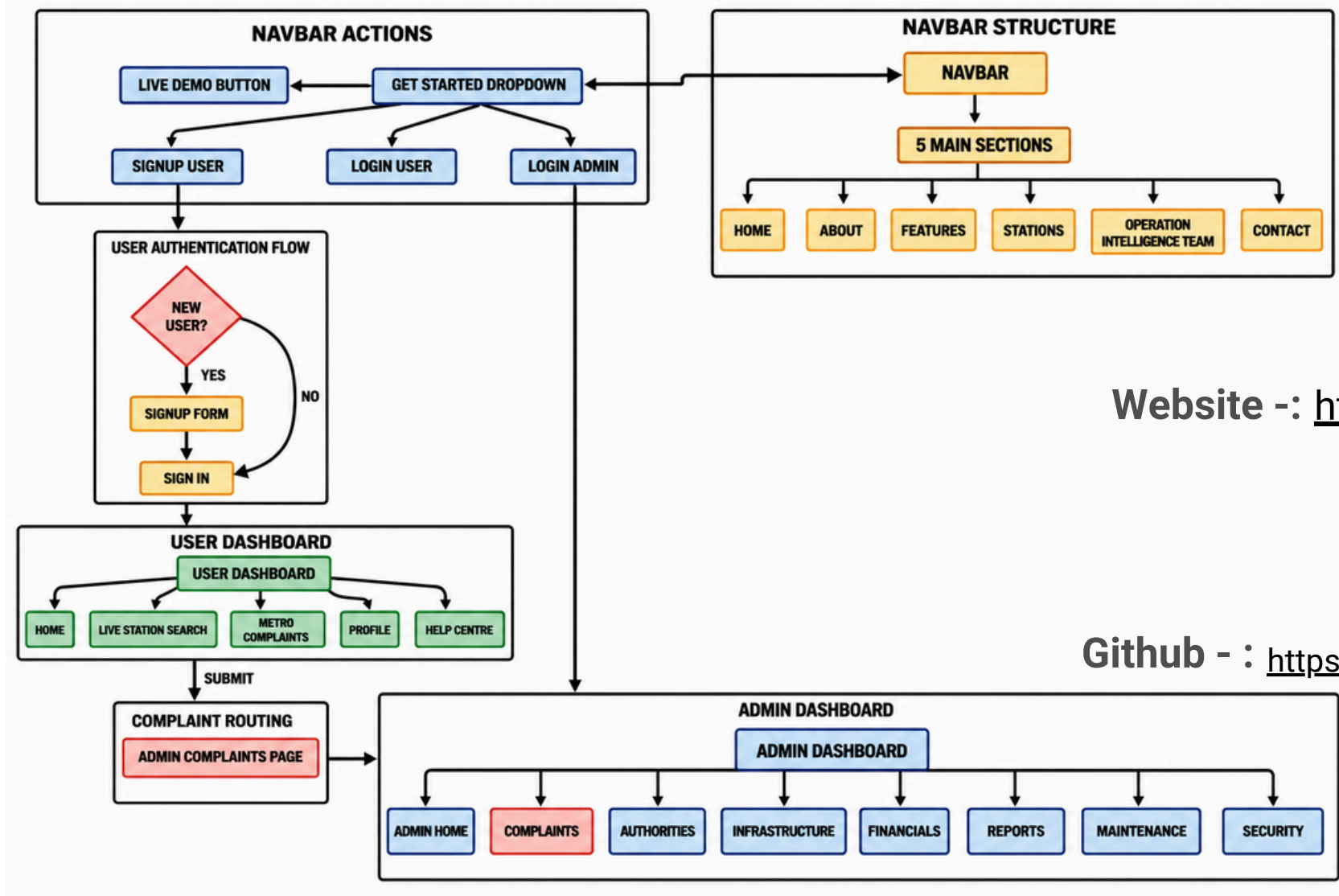
Deployment

Editor

Libraries



Live Demo / Prototype :-



Website - : <https://metro-mitra-rose.vercel.app/>



Github - : <https://github.com/Vibhav11Krishna/Metro-Mitra>

Feasibility & Practical Implementation :-



Social & Economic Impact:-

Transforming transport into a fast, clear, and resource-efficient utility.

- Social Benefit: Thousands of daily commuters gain a voice and faster resolution of travel issues.
- Economic Impact: Reduces resource waste by detecting duplicate complaints and automating manual tasks.
- Scalability: Modular architecture allows expansion from a pilot to a massive tool used across India.



Social Benefit

Empowers commuters and improves travel experience

Economic Impact

Reduces waste and automates processes

Scalability

Adapts to growing needs and widespread use

Model - U-Net (Encoder + Decoder) , Rest 50 Net

Future Enhancement & Conclusion

Image Segmentation

- Rapid reDetects and isolates the reported issue from uploaded images.pons to incident

Senior Citizen Assistance

- Dedicated support for elderly commuters with easy navigation and priority services.

Emergency SOS

- Rapid response to incident

Computer Vision/Open CV2



- Real-Time CCTV Analytics & Hazard Detection